Dallas AI - Agent Workshop

Mac Setup Guide

Detailed instructions for first time users

Use this guide if you are not comfortable coding or have limited prior coding experience using jupyter notebooks only

Before You Begin

This guide will walk you through every step of setting up the Dallas AI Agent Workshop on your Mac. By the end, you will have a working AI agent running on your laptop. The entire process takes about 20–30 minutes.

What You Will Need

- A Mac running macOS 12 (Monterey) or later
- A stable internet connection
- An OpenRouter account (free — we'll create one together)
- Optionally, a Tavily account for the Research Agent section (also free)

> **Tip: Check your macOS version**
> Click the Apple menu (□) in the top-left corner of your screen, then click "About This Mac." Your macOS version will be listed near the top.

Part 1: Opening Terminal

Terminal is a built-in Mac app that lets you type commands to control your computer. Think of it like sending text messages to your Mac — you type a command, press Return, and your Mac does what you asked.

Step 1: Open Terminal

There are two easy ways to open Terminal:

Option A — Spotlight Search: Press **Command + Space** on your keyboard. A search bar will appear. Type **Terminal** and press **Return**.

Option B — Finder: Open Finder, go to **Applications > Utilities**, and double-click **Terminal**.

Step 2: Get comfortable with Terminal

You should see a window with a blinking cursor. This is where you'll type commands throughout the workshop. You can copy commands from this guide and paste them into Terminal using **Command + V**.

> **Tip: Keep Terminal open**
> You'll use Terminal for the rest of this guide. Consider keeping it in your Dock by right-clicking the Terminal icon and selecting "Options > Keep in Dock."

Part 2: Installing Required Tools

Before we can run the workshop, we need to install a few tools. We'll install them one at a time.

2A: Install Homebrew (Mac's Package Manager)

Homebrew is a free tool that makes it easy to install software on your Mac. It's like an App Store for developer tools.

Step 1: Install Homebrew

Copy and paste the following command into Terminal, then press **Return**:

```
/bin/bash -c "$(curl -fsSL
  https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

You may be asked to enter your Mac password. When you type it, you won't see any characters appear — this is normal for security. Just type your password and press **Return**.

The installation may take a few minutes. Wait until you see the regular prompt again.

Step 2: Add Homebrew to your path (Apple Silicon Macs)

If your Mac has an Apple chip (M1, M2, M3, or M4), Homebrew may ask you to run two additional commands at the end of the installation. They will look something like this:

```
echo 'eval "$(/opt/homebrew/bin/brew shellenv)"' >> ~/.zprofile
eval "$(/opt/homebrew/bin/brew shellenv)"
```

If you see these commands in the Terminal output, copy and paste them exactly as shown, one at a time, pressing **Return** after each.

Step 3: Verify Homebrew is installed

Type this command and press **Return**:

```
brew --version
```

You should see a version number like "Homebrew 4.x.x". If you see "command not found," close Terminal, reopen it, and try again.

2B: Install Python

Python is the programming language the workshop uses. Your Mac may already have it, but we'll install a fresh version to be safe.

Step 1: Install Python with Homebrew

```
brew install python
```

This will download and install the latest version of Python.

Step 2: Verify Python is installed

```
python3 --version
```

You should see something like “Python 3.12.x” or similar. Any version 3.9 or higher is fine.

2C: Install Git

Git is a tool for downloading code from the internet. The workshop code is stored on a website called GitHub, and Git is what lets us download it.

Step 1: Install Git

```
brew install git
```

Step 2: Verify Git is installed

```
git --version
```

You should see a version number. If so, you’re good to go.

Part 3: Download the Workshop Code

Now we'll download the workshop files from GitHub to your Mac.

Step 1: Navigate to your Desktop

Type this command to make sure you're working from your Desktop:

```
cd ~/Desktop
```

Step 2: Download (clone) the workshop

Type this command to download the workshop files. Replace the URL below with the one provided by your workshop host:

```
git clone https://github.com/jiankunliu-ai/dallas-ai-agent-workshop
```

This creates a new folder on your Desktop containing all the workshop files.

Step 3: Move into the workshop folder

```
cd dallas-ai-agent-workshop
```

(Your host will confirm the exact folder name. It may differ slightly.)

> ⚠ **Important: Folder name**
> The folder name must match exactly. If you get an error like "no such file or directory," ask your host for the correct repository name.

Part 4: Set Up the Python Environment

We'll create an isolated workspace (called a virtual environment) so the workshop's software doesn't interfere with anything else on your Mac.

Step 1: Create the virtual environment

```
python3 -m venv .venv
```

This creates a hidden folder called **.venv** inside the workshop directory.

Step 2: Activate the virtual environment

```
source .venv/bin/activate
```

You'll know it worked if you see **(.venv)** at the beginning of your Terminal prompt. This means your isolated workspace is active.

Step 3: Install the workshop's required packages

```
pip install -r requirements.txt
```

This downloads all the Python packages the workshop needs. It may take a minute or two.

> **Tip: Always activate before working**
> Every time you open a new Terminal window to work on this workshop, you'll need to navigate to the folder and activate the environment again: cd ~/Desktop/dallas-ai-agent-workshop source .venv/bin/activate

Part 5: Get Your API Keys

The workshop uses AI models hosted online. You need a free API key to connect to them.

5A: OpenRouter API Key (Required)

OpenRouter gives you access to many AI models through a single service.

Step 1: Create an OpenRouter account

Open your web browser and go to:

https://openrouter.ai

Click **Sign Up** and create a free account. You can sign up with Google or email.

Step 2: Generate an API key

Once logged in, go to:

https://openrouter.ai/keys

Click **Create Key**. Give it any name (e.g., “workshop”). Copy the key that appears — it starts with **sk-or-**. Save it somewhere safe (like a note on your phone) — you won’t be able to see it again.

> ⚠ **Important: Keep your key secret**
> Your API key is like a password. Don’t share it publicly or post it online.

5B: Tavily API Key (Optional — for Research Agent)

Tavily provides web search capabilities for the research agent section. This is optional but recommended.

Step 1: Create a Tavily account

Open your browser and go to:

https://tavily.com

Click **Sign Up** and create a free account.

Step 2: Copy your API key

After signing up, you’ll find your API key on your dashboard. Copy it — it starts with **tvly-**.

Part 6: Configure Your Environment

Now we'll tell the workshop code about your API keys.

Step 1: Create your configuration file

Make sure you're in the workshop folder, then type:

```
cp .env.example .env
```

This creates a file called **.env** where your keys will be stored.

Step 2: Open the file for editing

Two ways

Using Mac Finder & TextEdit

Using Mac Finder go to folder: Desktop/dallas-ai-agent-workshop

- Press **Command + Shift + Period (** `⌘ + ⇧ + .` **)** simultaneously. This will enable showing 'hidden' files (files that begin with ".") like '.env' created above
- Open '.env' with 'TextEdit' (DO NOT USE word)
- Replace the text '`your_key_here`' with keys from openrouter & tavily. Leave `OPENROUTER_MODEL` as is

Command line *(skip if you completed step 2 using Mac Finder & TextEdit)*

Type this command to open the file in a simple text editor:

```
nano .env
```

You'll see something like:

```
OPENROUTER_API_KEY=your_key_here
OPENROUTER_MODEL=arcee-ai/trinity-large-preview:free
TAVILY_API_KEY=your_key_here
```

Enter your keys

Use the arrow keys to move to the line that says **OPENROUTER_API_KEY=**. Delete **your_key_here** and paste your OpenRouter key (**Command + V**).

If you have a Tavily key, do the same for the **TAVILY_API_KEY** line.

Leave the **OPENROUTER_MODEL** line as-is unless your host tells you otherwise.

Save and close the file

Press **Control + O** (the letter O, not zero) to save.

Press **Return** to confirm the filename.

Press **Control + X** to exit the editor.

> **Tip: Nano keyboard shortcuts**
> In the nano editor, the ^ symbol means the Control key. So ^O means “Control + O” and ^X means “Control + X.” The shortcuts are shown at the bottom of the editor.

Step 3

Type this command to open the file in a simple text editor:

```
pip install ipykernel
python -m ipykernel install --user --name=workshop --display-name
"Workshop (.venv)"
```

Part 7: Test Your Setup

Let’s make sure everything is working before the workshop begins.

Step 1: Run the preflight check

```
python test_model.py
```

Step 2: Check the result

If everything is configured correctly, you’ll see:

```
MODEL WORKING
```

If you see this — congratulations! You’re all set.

What If It Doesn’t Work?

Here are the most common issues and how to fix them:

Error	How to Fix
401 Error	Your API key is incorrect or expired. Go back to openrouter.ai/keys, create a new key, and update your .env file.
"command not found" for python	Try typing python3 instead of python, or re-run: brew install python
"No module named..."	Make sure your virtual environment is active. You should see (.venv) in your prompt. If not, run: source .venv/bin/activate
Notebook shows 401 but test_model.py works	You may have a stale environment variable. Before starting Jupyter, run: unset OPENROUTER_API_KEY OPENROUTER_MODEL

Part 8: Launch the Workshop Notebook

You're in the home stretch! Now we'll open the interactive notebook where the workshop takes place.

Step 1: Start Jupyter Lab

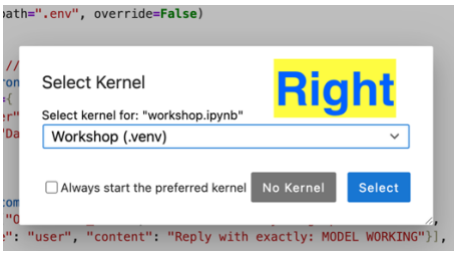
```
jupyter lab
```

Your web browser should automatically open to a page that looks like a code editor. If it doesn't open automatically, look in Terminal for a URL that starts with **http://localhost:8888** and paste it into your browser.

Step 2: Open the workshop notebook

In the file browser on the left side of Jupyter Lab, click on **workshop.ipynb** to open it.

Note: If you have anaconda, it may ask you to select kernal. Make sure to select "Workshop(.venv)"



Step 3: You're ready!

The workshop facilitator will guide you through the notebook from here. You can run each code cell by clicking on it and pressing **Shift + Return**.

> ⚠ **Don't close Terminal**
> Jupyter Lab runs inside Terminal. If you close the Terminal window, Jupyter will stop. Minimize it instead. When you're finished with the workshop, come back to Terminal and press Control + C to shut down Jupyter.

Quick Reference Card

Bookmark this page! Here's a summary of the key commands you'll use:

What You Want To Do	Command
Go to the workshop folder	`cd ~/Desktop/dallas-agent-workshop`
Activate the environment	`source .venv/bin/activate`
Run the connection test	`python test_model.py`
Start Jupyter Lab	`jupyter lab`
Stop Jupyter Lab	`Press Control + C in Terminal`
Edit your API keys	`nano .env`
Run a notebook cell	`Click the cell, then Shift + Return`

You're all set — enjoy the workshop! 🚀